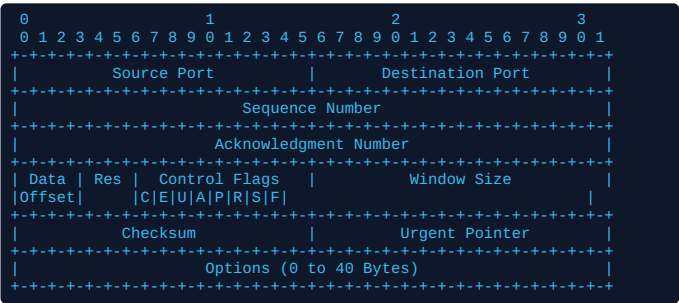


1. TCP PROTOCOL OVERVIEW

TCP operates at **Layer 4 (Transport Layer)** of the OSI Model [cite: 1]. It is connection-oriented, reliable, and guarantees in-order byte stream delivery [cite: 1]. Prepends a **20 to 60 Byte header** to data segments [cite: 1, 2].

2. TCP HEADER STRUCTURE (20–60 BYTES)



3. TCP HEADER FIELD BREAKDOWN

FIELD	BITS	FUNCTION
Source Port	16	Sender application port [cite: 3]
Dest Port	16	Receiver target port (e.g. 80, 443) [cite: 3]
SEQ Num	32	Tracks order of bytes sent [cite: 3]
ACK Num	32	Next byte expected from sender [cite: 3]
Data Offset	4	Header length in 32-bit words (5-15) [cite: 3]
Reserved	3	Set to zero (future use) [cite: 3]
Control Flags	9	Manage state & connection properties [cite: 3]
Window Size	16	Receiver available buffer size [cite: 3]
Checksum	16	Error checking for header & payload [cite: 3]
Urgent Ptr	16	End of urgent data (if URG active) [cite: 3]
Options	0–40B	MSS, Window Scale, Timestamps [cite: 3]

4. CONTROL FLAGS BREAKDOWN (9 BITS)

FLAG	DESCRIPTION & FUNCTIONALITY
NS	Nonce Sum: ECN concealment protection [cite: 4]
CWR	Congestion Window Reduced [cite: 4]
ECE	ECN-Echo: Network congestion experienced [cite: 4]
URG	Urgent Pointer field is active [cite: 4]
ACK	Acknowledgment Number is valid [cite: 4]
PSH	Push data immediately to app buffer [cite: 4]
RST	Reset / abort unstable connection [cite: 4]
SYN	Synchronize sequence numbers (Handshake) [cite: 4]
FIN	Finish connection termination request [cite: 4]

5. KEY PERFORMANCE METRICS

5.1 Sequence Number (SEQ) — Layer 4

32-bit tracking number for first byte of payload [cite: 5]. Enables packet reordering and detects missing bytes for retransmission [cite: 5, 6].

5.2 Maximum Transmission Unit (MTU) — Layer 3

Max frame size an interface transmits without fragmentation [cite: 7]. Standard Ethernet = **1500 Bytes** [cite: 8].

MTU = IP Header + TCP Header + TCP Payload (MSS) [cite: 9]

5.3 Maximum Segment Size (MSS) — Layer 4

Max payload capacity per segment (excluding headers) [cite: 10]. Exchanged via Options in SYN handshake [cite: 11].

MSS = MTU - (IP Header [20B] + TCP Header [20B]) [cite: 12]
Standard MSS = 1500 - 40 = 1460 Bytes [cite: 13]

5.4 Window Size (WND) & Flow Control — Layer 4

Buffer space receiver advertises to prevent fast senders from overwhelming it [cite: 14, 15].

Packets per Burst = Window Size / MSS [cite: 16]
Ex: 58,400 B / 1,460 B = 40 Packets [cite: 17, 18, 19]

5.5 Round-Trip Time (RTT) — End-to-End

Total time for a packet to reach destination and ACK to return [cite: 20].

RTT = Propagation + Processing + Queuing + Transmission [cite: 21]

- **Dynamic Timeout (RTO):** Used to calculate retransmission timeouts [cite: 22].
- **Congestion Avoidance:** Spikes warn of router buffer bloat [cite: 23].

6. METRIC REFERENCE TABLE

METRIC	LAYER	UNIT	TYPICAL	PRIMARY FUNCTION
SEQ Num	Layer 4	Byte	32-bit	Reordering & delivery [cite: 24]
MTU	Layer 3	Bytes	1500 B	Physical frame limit [cite: 24]
MSS	Layer 4	Bytes	1460 B	Max payload capacity [cite: 24]
Window	Layer 4	Bytes	65,535 B	Flow control buffer [cite: 24]
RTT	Path	ms	10–150ms	Path latency evaluation [cite: 24]